Negative numbers, coordinates, and inequalities

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. 10×3

2. 10×7

3. 10×4

4. 10×9

5. A parallelogram has base 6, height 8, and slanted side 10 cm. Find its area.

6. 9 of 12 tiles are green. What percent are green?

7. 4×6

8. Find $|5|$ and describe it as a distance.

9. Order -4, 0, -9 from least to greatest.

10. Find $|-12|$ and describe it as a distance.

11. Find $|0|$ and describe it as a distance.

12. Find the opposite of -6.

13. Ben says $-10 > -3$ because $10 > 3$. Use a number line to explain and repair his statement.

14. Write signed numbers for 5 meters below sea level and 4 meters above it. State what zero represents.

15. Order these values from least to greatest: $\frac{3}{2}$, -5, 0, -2.

16. Find the opposite of -5, then its opposite. Explain why you return to the start.

17. Find $|-5|$, $|5|$, and $|0|$. Compare the first two.

18. Which is warmer, $-5°C$ or $-7°C$? Which has greater absolute value?

CHALLENGE / Explain your plan and check the result

C1. A trail marker is at -7 km on an east-west line. Another is its opposite. Find the distance between them. A shelter at -2 km lies between them; find both distances from the shelter and check that they add to the whole distance.

C2. Find a negative number that is 10 units from 3. Then find its opposite. Compare the two numbers and their absolute values. Explain why the larger number does not have the larger absolute value.